



Customer Behavior Analysis

Revenue drivers, customer segments & discount effectiveness — analyzed with **Python, SQL, and Power BI**

METHODOLOGY

Three-Phase Approach



Python EDA

Cleaning, distributions & feature engineering



SQL Analysis

Segmentation, aggregation & behavioral metrics



Power BI

4 interactive dashboards with filters



DATASET

3,900 Records · 18 Columns

Demographics

Age, Gender, Location

Transactional

Purchase amount, items, frequency

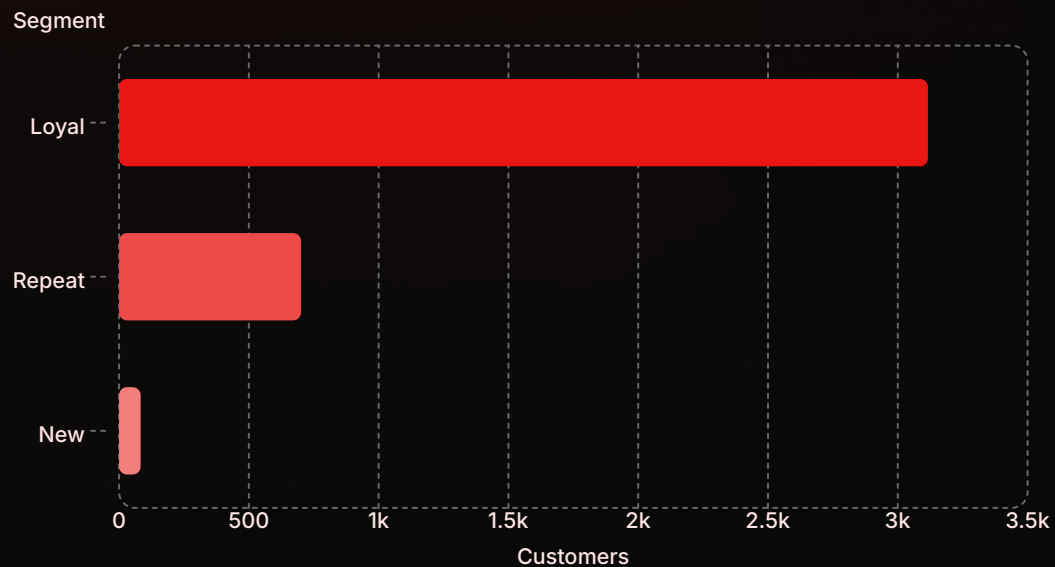
Marketing

Discounts, subscriptions, promo codes

Satisfaction

Review ratings (37 nulls filled)

Revenue & Segmentation



Key Findings

- **Loyal customers** = 85% of base
- **Male revenue:** \$157K vs Female \$75K
- **Young Adults** lead revenue: \$62K
- **Subscribers:** 1,053 customers, \$62K revenue

POWER BI

Overview Dashboard

\$233K

Total Revenue

3.9K

Total Customers

\$59.76

Avg Order Value

97.9%

Repeat Purchase Rate

Clothing dominates at ~45% of revenue. Young Adult & Middle-Aged are core segments. Payment/shipping show **minimal impact**.



CUSTOMER BEHAVIOR

Loyalty Drives Everything

85% Loyal Customer Rate

~\$199K revenue from loyal segment alone

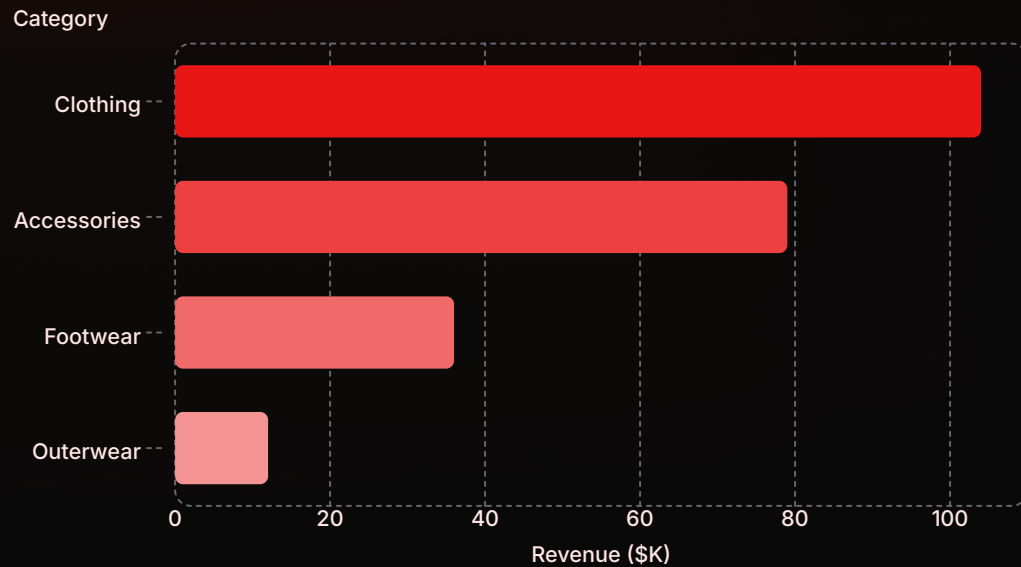
AOV Consistent Across Segments

Loyal \$59.73 · Repeat \$60.19 · New \$35.47

Structured Buying Cycles

Clusters at 14-day & 90-day intervals — timing opportunity

Where Revenue Comes From



Takeaways

- **Clothing = 45%** — strong but risky concentration
- Revenue **evenly distributed** across locations
- **Minimal seasonal variation** — stable year-round demand
- Growth via **product mix & cross-selling**, not seasonality

DISCOUNT IMPACT

Discounts: Limited Effectiveness



Key Signals

- Only **43%** usage — moderate adoption
- AOV **-1.42%** with discounts — margin trade-off
- Customer mix **unchanged** by discount use
- **No female customers** used discounts — data anomaly



RECOMMENDATIONS

Three Strategic Priorities



Optimize Discounts

Target new/low-frequency buyers only — cut broad promotions



Strengthen Loyalty

Non-price incentives; trigger engagement at 14/90-day cycles



Expand Core Categories

Invest in Clothing; cross-sell to Young Adult & Middle-Aged

Read With Caution

SKEWED LOYALTY



Dataset heavily skewed toward loyal customers (97.9% repeat rate likely inflated).

GENDER DATA GAP



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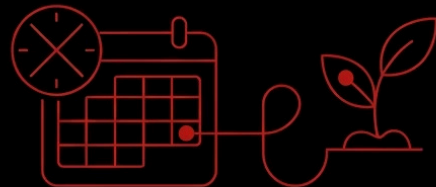
No female discount usage, suggesting data quality issue.

LIMITED VARIABILITY



Low variance in AOV, geography, and seasonality observed.

MISSING LIFECYCLE



No time-based or cohort data for lifecycle analysis.

Implications

- Retention insights may be **overestimated**
- Validate anomalies before strategic decisions
- Findings are **directional**, not definitive
- Future work: cohort & time-series analysis